GG-L-260D
December 30, 1980
SUPERSEDING
Fed. Spec. GG-L-260C
August 6, 1975

FEDERAL SPECIFICATION

LEVELS, SURVEYING: ABNEY TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers Abney type hand levels.

1.2 Classification.

1.2.1 Types. The levels shall be the following types, as specified (see 6.2):

Type I - Level, glass prism reflector.
Type II - Level, polished chromium plated metal reflector.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

DD-G-451 - Glass, Float or Plate, Sheet, Figured (Flat, For Glazing, Mirrors and Other Uses).

Federal Standards:

Fed. Std. No. H28 - Screw-Thread Standards For Federal Services.
Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

FSC 6675

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(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

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Military Specification:

MIL-S-3534	- Surveying Instruments and Accessories, Packaging of.

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables For Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The level shall consist of a square metal sighting tube, eyepiece, index arm, focusing slide, half lens, sighting reference (horizontal cross hair), arc and arc plates, scale index, bubble vial, neck strap stud or ring, bubble vial adjusting tool if needed, and a case.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

3.2.1 Metal. All metal parts shall be brass or aluminum except the arcs, which shall be corrosion resistant steel, aluminum, or nickel silver.

3.2.2 Glass. The glass used in the reflector shall conform to DD-G-451, type I, class 1, quality q1, when applicable (see 3.6).

3.3 Screw threads. All screws and screw threads shall be in accordance with Fed. Std. No. H28 and have a class 2 fit.

3.4 Environmental conditions.

3.4.1 Operating temperature. The level shall perform as specified herein in any ambient temperature from plus 110° F (plus 43.3° C) to minus 25° F (minus 32° C).

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3.4.2 Storage temperature. The level shall not be damaged or lose calibration when stored up to 6 hours under any ambient temperature conditions within the range of plus 155° F (plus 68° C) to minus 30° F (minus 34.4° C). The level shall withstand the tests of sections 4.4.2.1 and 4.4.2.2 without failure.

3.4.3 Vibration. When tested in accordance with 4.4.2.3, the level shall show no signs of physical damage or loosening of any parts, including fasteners.

3.5 Eyepiece. The eyepiece shall be circular in shape without any exposed sharp edges and shall be either a part of, or securely attached to, the level sighting tube or focusing draw tube. If it is a separate piece, attachment shall be by soldering or other equally strong joining method. The eyepiece aperture shall include the entire field of view of the tube and bring the sighting reference into sharp focus.

3.6 Reflector. The reflector shall be securely mounted on a removable metal sleeve or plate.

3.7 Arc plate and arcs. The outside face of the arc plate shall have an arcual (curved-like) recess to receive two arcs at the same time. The arc plate shall be directly and immovably mounted on the sighting tube and shall support the bubble vial bracket with an interconnected moving index arm for reading the arcs. Each arc shall have graduated scales with uniformly etched black lines and numbers. One arc shall read degrees on the front side and percent on the reverse side; the other arc shall read topographic on the front side and may read percent, chainage correction, or be blank on the reverse side. The two arcs shall be held in place by two thumb screws with approximate 0.375-inch (9.525 millimeter (mm)) diameter knurled heads. The screws shall have a minimum engagement of 3 threads. The scale graduations shall be in accordance with table I.

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TABLE I. Scale graduations

Scale	Range	Lines	Numbers
Degree	60 degrees above or below horizontal	Each degree. Fifth degree line longer.	Each 10 degrees.
Percent	150 percent above or below horizontal	Each percent, 0 to 60. Fifth percent line longer. Each 2 percent, 60 to 120. Ten percent line longer. Each 5 percent, 120 to 150. Ten percent line longer.	Each 10 percent, 0 to 60. Each 20 percent, 60 to 120. At 150 percent.
Topographic	100 above or below horizontal	Each unit, 1 to 45. Fifth unit line longer. Each 5 units, 45 to 70. Each 10 units, 70 to 100.	Each 10 units, 0 to 80. At 100.
Chainage Correction	0 to 25 above or below horizontal	Each 1/10 unit, 0 to 1. 0.5 unit longer. Each half unit, 2 to 15. Each unit, 15 to 25.	First unit. Each 2 units, 2 to 8. Each 5 units, 10 to 25.

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3.8 Bubble vial. The bubble vial shall be glass or plastic and it shall be enclosed as needed to prevent breakage. The bubble image shall be spherical in shape and shall be fully visible in the reflector at all inclinations from minus 60° to plus 60° from the horizontal when the bubble image is bisected by the index line. The vial bubble shall have a sensitivity of not less than 2 mm of movement for 20 minutes of change in the arc index arm position. Bubble sensitivity shall be determined by the test in 4.4.2.5. The shape of the vial shall keep distortion of the spherical bubble image to a minimum throughout the full range of operation of the level. The liquid in the vial shall have a freezing point below minus 65° F (minus 54° C) and a boiling point above plus 155° F (plus 68° C). All alignment and calibration adjustments shall be made by the manufacturer or contractor before delivery.

3.9 Index arm. The level shall have a movable index arm with a lock for reading arcs and shall have a mechanical or friction drive for fine adjustment of the slope angle. There shall be no slippage of the fine adjustment. The index arm shall be flush with the arc being used, shall be immediately adjacent to the graduated arc scale, and shall be usable over the full operating range of any arc provided under this specification. No adjustment of the bubble vial shall be necessary when changing scales.

3.10 Bubble index line. A bubble index line shall be provided to indicate the desired bubble position (mid-point of the vial) when the vial is in a true horizontal plane. If the index provided is a line visible in the level's reflector, it shall be scribed on the vial and shall be as close to the pivot axis of the index arm as possible.

3.11 Sighting tube and sighting reference.

3.11.1 Sighting tube. The sighting tube shall be not less than 6 inches (152.4 mm) nor more than 8 inches (203 mm) in length and 0.65 to 0.75 inches (16.5 to 19 mm) square. The interior of the sighting tube shall be coated or treated with a black nonreflective finish.

3.11.2 Sighting reference. The sighting reference may be on the sighting tube or on the removable metal sleeve. It shall be securely attached by soldering or brazing. In addition it shall be recessed approximately 0.125 inch (3.175 mm) from the front of the sighting tube.

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3.12 Focusing. The level shall be provided with a means for bringing the vial bubble image into sharp focus with the horizontal sighting reference as viewed by the level user.

3.13 Accuracy. The total level error shall not be more than 0.50 degree or its equivalent on either arc.

3.14 Case. A case made of leather, artificial leather, or plastic shall be supplied to protect the level when not in use. The material used shall be not less than 0.047 inch (1.194 mm) thick. The case shall have a flap with a fastener to keep the cover closed. A belt strap shall be provided on the back of the case near the top for fastening the case to the user's belt. The strap shall be at least 1.5 inches (38 mm) wide and form a loop. The opening in the loop shall be formed to fit a belt 0.125 inch (3.175 mm) thick by 2 inches (50.8 mm) wide. The belt strap shall be securely sewn or riveted to the case with brass or enamel-coated rivets. Provision to securely fasten special tools for adjusting bubble vial shall be provided inside the case.

3.15 Neck strap attachment. A stud with a hole or ring shall be provided for attaching a neck strap.

3.16 Identification marking. The manufacturer's name or trademark shall be stamped or otherwise permanently affixed on each item.

3.17 Treatment and painting. Unless otherwise specified (see 6.2), the portions of the level normally painted shall be cleaned, treated, and painted in accordance with the manufacturer's standard practice.

3.18 Type I level. Type I level shall contain a glass prism reflector.

3.19 Type II level. Type II level shall contain a polished chromium plated metal reflector.

3.20 Workmanship. All parts of the level shall be clean and free from any harmful extraneous material. All exposed edges shall be rounded or beveled. All fittings shall be securely mounted and all screws shall be fully turned down. Movable parts shall move smoothly and positively throughout the required range.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level S-2. The lot size shall be all of the completed levels offered for inspection.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The levels shall be examined for the following defects:

101. Materials not as specified.
102. Components missing or not as specified.

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103. Focusing slide does not bring the reflected bubble image into sharp focus.
104. Scale graduations and numbers not clear or uniform.
105. Bubble image not visible in the reflector when tube is inclined and vial level.
106. Sighting tube not as specified.
107. Focusing slide not as specified.
108. Identification marking not as specified.
109. Workmanship not as specified.

4.4.2 Tests.

4.4.2.1 Low temperature. The level shall be at a temperature of plus 68° F plus or minus 2° F (plus 20° C plus or minus 1° C) and a relative humidity of 50 percent plus or minus 5%. Place directly in a temperature controlled chamber at minus 25° F plus or minus 2° F (minus 32° C plus or minus 1° C) for one hour or until cold equilibrium is reached. All movements shall then be checked. Inability of the level to focus, or movable parts that bind or move loosely throughout their range of movement, shall constitute failure of this test. If the level operates satisfactorily, remove the level from the cold chamber to plus 68° F plus or minus 2° F (plus 20° C plus or minus 1° C) with 50 percent plus or minus 5% relative humidity until thermal equilibrium is reached. Recycle in the above manner for a total of 5 cycles. Replace the level in the temperature controlled chamber and gradually lower the temperature over a two-hour period to minus 30° F plus or minus 5° F (minus 34° C plus or minus 3° C) for a minimum of 24 hours. The focus and movable parts shall again be checked. Evidence of damage, malfunction, or loose calibration shall constitute failure of this test. When specified (see 6.2), the contractor may certify compliance in lieu of testing.

4.4.2.2 High temperature. From an ambient temperature of plus 68° F plus or minus 2° F (plus 20° C plus or minus 1° F) and 50 percent plus or minus 5% relative humidity, place the level into a hot chamber at plus 115° F plus or minus 2° F (plus 46° F plus or minus 1° C) for one hour or until thermal equilibrium is reached. All movements shall then be checked. Inability of the level to focus, or movable parts that bind or move loosely throughout their range of movement shall constitute failure of this test. If the level operates satisfactorily, then remove the level from the hot chamber to plus 68° F plus or minus 2° F (plus 20° plus or

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minus 1° F) with 50 percent plus or minus 5% relative humidity until thermal equilibrium is reached. Recycle in the above manner for a total of 5 cycles. Replace the level in the hot chamber and gradually raise the temperature to plus 155° F plus or minus 2° F (plus 68° C plus or minus 1° C) and maintain at the temperature for four hours. The temperature shall be lowered to room temperature over a period of 4 to 4.5 hours. The focus and movable parts shall again be checked. Evidence of damage, malfunction, or loose calibration shall constitute failure of this test. When specified (see 6.2), the contractor may certify compliance in lieu of testing.

4.4.2.3 Vibration test. The level shall be vibrated at a frequency of 30 cycles per second and an amplitude of 0.0625 inch (1.588 mm) for a period of 15 minutes. The index arm lock and all screws shall be tightened before testing. Any evidence of breakage or permanent deformation of any part shall constitute failure of this test.

4.4.2.4 Accuracy test. The level shall be placed in a horizontal position with the arc plate in a vertical plane on a test bench. The index shall be adjusted so that the bubble is centered on the sighting reference; the index shall then be read. The level shall be placed in an inclined position so that the line of sight is alternately plus 30°, minus 30°, plus 45°, minus 45°, plus 60°, and minus 60° from the horizontal. The index shall be read at each angle. Evidence of error of more than 0.5 degree or its equivalent on either arc shall constitute failure of this test.

4.4.2.5 Vial sensitivity. The vial shall be mounted on a smooth horizontal beam. One end of the beam shall be supported on a pivot and the other by a micrometer screw. The beam shall be calibrated so that each revolution of the micrometer screw will result in a known angular displacement of the beam, both above and below the horizontal plane, to within one minute of arc. The vial bubble movement shall be measured with a scale for movements of the beam both above and below the horizontal plane. A sensitivity of less than 2 mm of bubble movement per 20 minutes of arc movement of the beam in either direction from horizontal shall constitute failure of this test. When specified (see 6.2), the contractor may certify compliance in lieu of testing.

4.5 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be examined to determine compliance with the Quality Assurance provisions of MIL-S-3534.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, level B, or commercial, as specified (see 6.2).

5.1.1 Level A. Each level shall be preserved and packaged in accordance with the level A requirements of MIL-S-3534.

5.1.2 Level B. Each level shall be preserved and packaged in accordance with level B requirements of MIL-S-3534.

5.1.3 Commercial. The levels shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the levels against damage during multiple shipments, handling and storage.

5.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A. Levels preserved and packaged as specified in 5.1 shall be packed in accordance with level A requirements of MIL-S-3534.

5.2.2 Level B. Levels preserved and packaged as specified in 5.1 shall be packed in accordance with level B requirements of MIL-S-3534.

5.2.3 Commercial. The levels, preserved and packaged as specified in 5.1, shall be packed in fiberboard boxes to insure delivery at destination; provide for redistribution by the initial receiving activity; and be acceptable by common carrier under the National Motor Freight Classification and Uniform Freight Classification rules.

5.3 Marking.

5.3.1 Military agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

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6. NOTES

6.1 Intended use. The level is used for measuring the degree of land slope and the percent of grade, for topographic surveying, for staking lines in level contour or on uniform grades, and for chaining on uniform grades.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type level required (see 1.2).
(c) Treatment and painting required if other than as specified (see 3.17).
(d) When a certificate of compliance may be submitted in lieu of testing (see 4.4.2.1, 4.4.2.2, and 4.4.2.5).
(e) Degree of preservation, packaging and packing required (see 5.1 and 5.2).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

MILITARY INTERESTS:

Custodians:

Army - ME
Navy - YD

Review activity:

DLA - GS

User activities:

Army - CE
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
USDA - AFS

Preparing activity:

Army - ME

Project 6675-0698

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